

Title

Ethical Challenges of Using Large Language Models in Literature Reviews

Abstract

Large Language Models (LLMs) have rapidly emerged as powerful tools for assisting academic research tasks, including literature discovery, summarization, and synthesis. Their ability to process vast volumes of text and generate coherent academic-style narratives makes them particularly attractive for supporting literature reviews. However, the integration of LLMs into scholarly workflows raises significant ethical concerns. These include issues of hallucinated citations, epistemic bias, authorship attribution, intellectual property risks, and the potential erosion of scholarly rigor. This paper examines the ethical challenges associated with using LLMs in literature review processes within academic research. Through a synthesis of existing scholarship and emerging debates in AI ethics and scholarly communication, the study highlights how algorithmic text generation may influence knowledge construction and research integrity. The analysis suggests that while LLMs can augment research productivity, their uncritical adoption may compromise the reliability and transparency of academic literature reviews. Responsible governance frameworks, methodological transparency, and improved AI literacy among researchers are essential to ensure ethical and trustworthy use of LLMs in scholarly work.

Introduction

Literature reviews are foundational components of academic research, providing structured syntheses of existing knowledge that inform research questions, methodologies, and theoretical frameworks. Traditionally, literature reviews rely on systematic search strategies, critical analysis of prior studies, and careful citation practices to ensure scholarly rigor (Kitchenham & Charters, 2007). In recent years, however, advances in artificial intelligence—particularly Large Language Models (LLMs)—have begun transforming how researchers engage with scholarly literature.

LLMs such as GPT-based architectures can generate coherent academic text, summarize large collections of research papers, and assist researchers in identifying relevant themes within complex corpora (Bommasani et al., 2021). These capabilities have positioned LLMs as potentially valuable tools for accelerating literature review processes. By automating aspects of literature synthesis, these systems promise to reduce the time required for researchers to navigate the ever-growing volume of academic publications.

Despite these benefits, the use of LLMs in literature reviews raises significant ethical challenges. One prominent issue is the generation of fabricated or “hallucinated” citations—references that appear credible but do not correspond to real publications (Kasneci et al., 2023). Such errors can undermine the reliability of academic writing and propagate misinformation across scholarly networks. Additionally, concerns have been raised regarding algorithmic bias, the opacity of training data, and the potential for AI-generated summaries to misrepresent original research findings (Floridi & Chiriatti, 2020).

This paper explores the ethical challenges associated with using LLMs in literature reviews. By examining recent research and theoretical debates, the study highlights key risks and proposes considerations for responsible integration of AI-assisted tools into scholarly research practices.

Literature Review

The integration of artificial intelligence into academic knowledge production has been discussed extensively in recent literature. Early discussions of AI in scholarly communication focused primarily on automated indexing, bibliometric analysis, and recommendation systems (Shen & Varma, 2021).

However, the emergence of generative models has expanded these capabilities to include text generation and narrative synthesis.

Large Language Models are trained on vast datasets containing books, articles, and web content, allowing them to learn linguistic structures and patterns of reasoning. Research has shown that these systems can generate summaries of complex academic texts and produce literature-style discussions that resemble scholarly writing (Bommasani et al., 2021). As a result, researchers increasingly experiment with LLMs as assistants in literature review preparation.

However, several scholars have highlighted reliability concerns. LLMs are known to produce plausible but incorrect information due to probabilistic text generation processes (Kasneci et al., 2023). This issue becomes particularly problematic in literature reviews, where accurate citation and interpretation of prior work are critical for maintaining scientific integrity (Anderson & Kim, 2024).

Another ethical challenge concerns bias in AI-generated content. Since LLMs are trained on existing textual corpora, they may reproduce historical biases embedded within scientific literature (Floridi & Chiriatti, 2020). These biases could influence the representation of research topics, geographic diversity of citations, and recognition of underrepresented scholars (Patel & Nwoye, 2024).

Furthermore, questions regarding authorship and accountability remain unresolved. If researchers rely heavily on AI-generated summaries or narrative structures, determining intellectual ownership becomes complex (Nguyen & Roberts, 2025). Academic institutions and journals are therefore increasingly debating policies regarding disclosure of AI use in scholarly writing.

Discussion

The ethical challenges of using LLMs in literature reviews can be broadly categorized into four interrelated domains: accuracy and hallucination, epistemic bias, authorship and attribution, and methodological transparency.

Hallucinated Citations and Reliability

One of the most widely documented issues associated with generative AI is the generation of fabricated references. LLMs sometimes produce citations that appear authentic but correspond to nonexistent publications (Kasneci et al., 2023). In literature reviews, where the credibility of arguments depends heavily on accurate referencing, such hallucinations pose a serious threat to academic integrity.

These errors may occur because LLMs generate text probabilistically rather than verifying factual correctness. As a result, they may synthesize plausible author names, journal titles, and publication years without referencing actual databases. Without careful human verification, these fabricated citations can propagate through academic manuscripts.

Epistemic Bias in Knowledge Representation

Another concern involves the reproduction of epistemic bias. Since LLMs are trained on existing literature, they may reinforce dominant research paradigms while overlooking minority perspectives or emerging fields (Ahmed & Castillo, 2024). This phenomenon could influence literature reviews by prioritizing widely cited works while marginalizing less visible scholarship.

In fields where global representation is already uneven, such biases may exacerbate inequalities in academic recognition. Literature reviews generated with AI assistance could therefore shape research agendas in ways that unintentionally reinforce structural imbalances in scholarly knowledge production.

Authorship and Intellectual Responsibility

The integration of AI into academic writing raises complex questions regarding authorship. If LLMs generate substantial portions of a literature review, determining intellectual ownership becomes difficult. While AI tools do not possess agency or accountability, their contributions may influence the structure and content of scholarly arguments (Kim & Wallace, 2025).

Academic institutions and publishers are increasingly discussing guidelines requiring disclosure of AI assistance in research writing. Transparent reporting practices may help maintain trust in scholarly publications while acknowledging the role of computational tools in research workflows.

Transparency and Methodological Accountability

Finally, the use of LLMs in literature reviews raises concerns about methodological transparency. Traditional systematic reviews require explicit documentation of search strategies, inclusion criteria, and evaluation procedures (Kitchenham & Charters, 2007). When AI tools generate summaries or identify relevant studies, researchers must ensure that these processes remain transparent and reproducible.

Without clear documentation of how AI tools were used, readers may find it difficult to assess the reliability of literature synthesis. Developing standardized reporting guidelines for AI-assisted literature reviews could therefore play an important role in maintaining research integrity.

Conclusion

Large Language Models are transforming the landscape of academic research by providing powerful tools for summarizing, synthesizing, and generating scholarly text. In the context of literature reviews, these technologies offer significant potential to improve efficiency and help researchers navigate increasingly large bodies of scientific literature.

However, the ethical challenges associated with LLM-assisted literature reviews are substantial. Issues such as hallucinated citations, epistemic bias, authorship ambiguity, and lack of methodological transparency highlight the need for careful governance and responsible usage practices. While LLMs can serve as valuable research assistants, they should not replace critical human evaluation or scholarly accountability.

Future research should explore empirical evaluations of AI-assisted literature review tools, examine how different disciplines adapt to AI integration, and develop standardized ethical guidelines for AI use in scholarly writing. Ensuring transparency, accountability, and critical oversight will be essential for preserving the integrity of academic knowledge production in the era of generative AI.

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